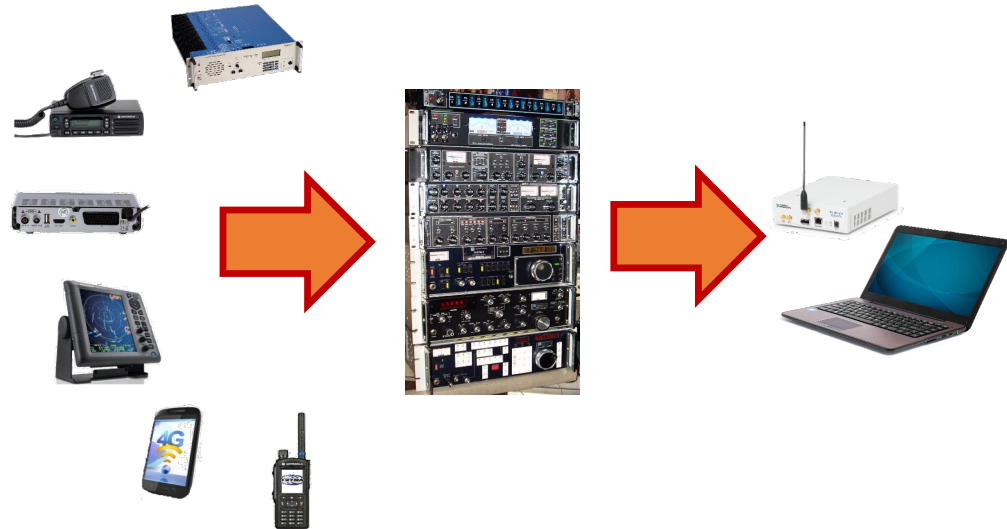


Software defined radio

SDR concept

- In conventional radios, functions such as modulation, demodulation, filtering and frequency conversion are implemented by dedicated analog or digital hardware.
- Advances in digital processors and analog-to-digital converters allow an increasing fraction of these functions to be implemented in software.



From the Complex Envelope to Software-Defined Radio

- Passband signals are represented through their complex envelope:

$$s(t) = \text{Re}\{\tilde{s}(t)e^{j2\pi f_c t}\}$$

- Let $v(t)$ denote the received passband signal. Its complex envelope $\tilde{v}(t)$ contains the amplitude and phase evolution of the received RF signal.

$$\boxed{\tilde{s}(t) \rightarrow s(t) \rightarrow \text{physical channel} \rightarrow v(t) \rightarrow \tilde{v}(t)}$$

- A software-defined radio converts a selected portion of the received RF spectrum into a sequence of complex-baseband samples

$$\tilde{v}[k] = v_I[k] + jv_Q[k]$$

SDR paradigm

- In an SDR receiver the incoming signal is converted to a digital format and then the signal is processed digitally.
- The RF front end is *software-configurable*, while baseband processing is implemented digitally and can be reprogrammed.
- A common hardware platform can support different modulation formats, algorithms, products and applications through software reconfiguration, reducing development and production costs.

SDR – Historical milestones

1984 - E-Systems Inc. (Raytheon) coined the term 'software radio'

1992 - J.Mitola publishes the paper: 'Software Radios: Survey, Critical Evaluation and Future Directions',



2010–2012 — The possibility of extracting raw 8-bit I/Q samples from RTL2832U-based DVB-T dongles led to the RTL-SDR open-source project.

1984

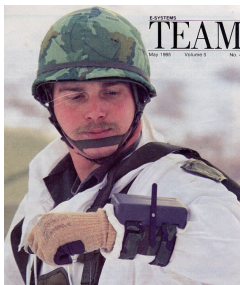
1991

1992

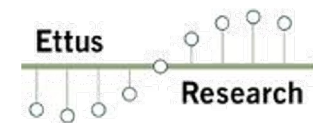
2005

2011

SPEAKeasy



1991 - SPEAKeasy is the first military program that specifically required a radio to have its physical layer components. The objective was a single radio that could support ten different military radio protocols



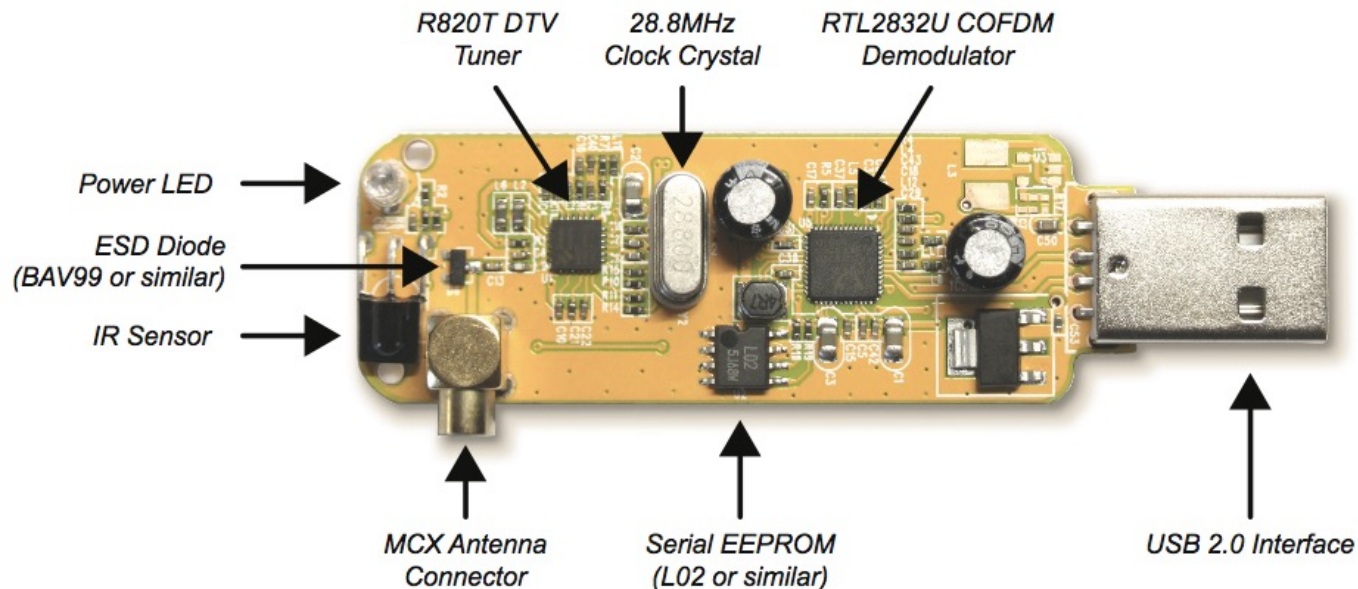
2005 - Ettus Research introduced the USRP1, one of the first widely accessible general-purpose SDR platforms.

The RTL-SDR Platform

- The SDR paradigm will be explored using an RTL-SDR, a low-cost, receive-only software-defined radio based on USB DVB-T dongles.
- It contains a tunable RF front end and a digital processing section.
- The selected RF band is converted into complex baseband I/Q samples:
$$\tilde{v}[k] = v_I[k] + jv_Q[k].$$
- The samples are transferred to a host computer through USB and processed in software.
- The RTL-SDR can be tuned over a wide RF range, but it captures only a limited bandwidth around the selected center frequency.

RTL-SDR architecture

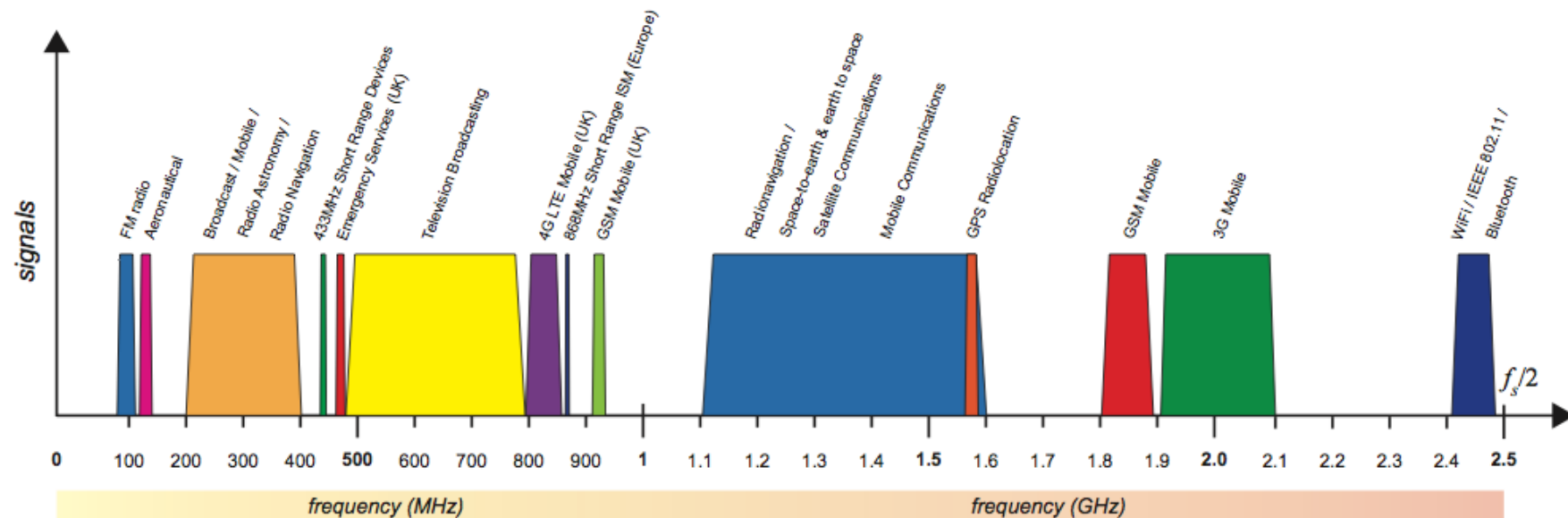
From RF signals to digital I/Q samples



The analog RF tuner selects and downconverts the desired signal, while the RTL2832U digitizes it and generates complex baseband I/Q samples.

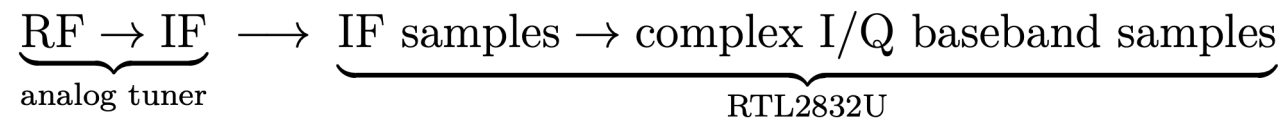
Representative Radio Services in the 0.1–2.5 GHz Range (UK chart)

- Since the RTL-SDR can be tuned over a wide RF range, it can be used to observe signals belonging to many different radio services.
- Actual reception depends on the tuner range, antenna, signal bandwidth, received power and signal format.



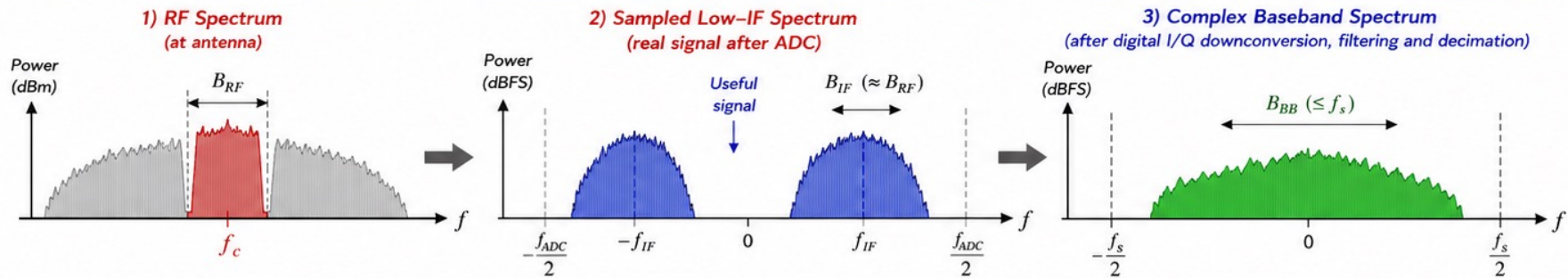
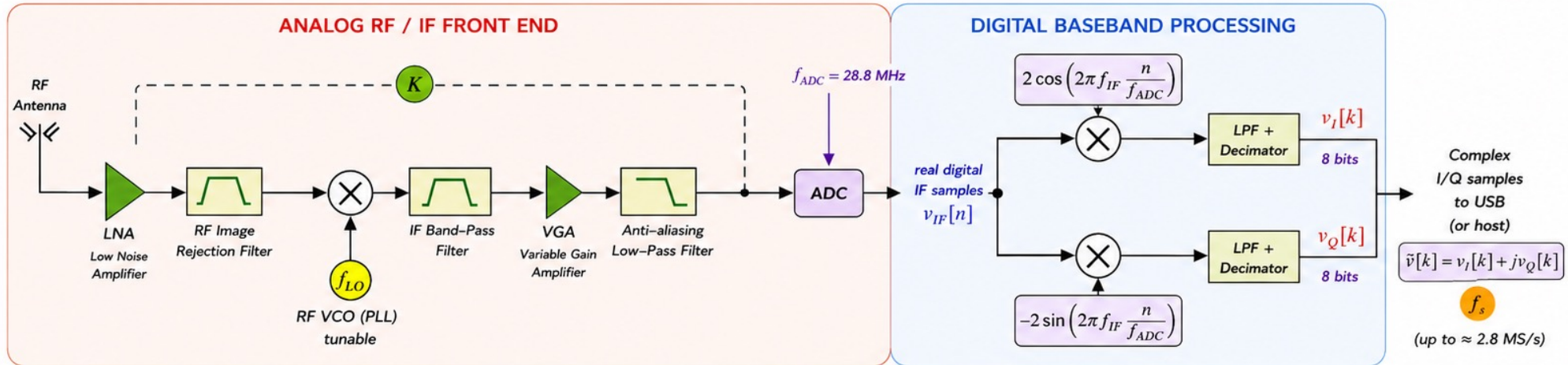
From RF to Complex Baseband in Two Stages

- The RTL-SDR contains two main signal-processing stages:
 - the analog tuner selects the desired RF band and translates it to a lower *intermediate* frequency (IF);
 - the RTL2832U samples the IF signal and performs digital quadrature downconversion, filtering and decimation.



- The analog tuner relies on *heterodyne conversion* to translate the selected RF band to an intermediate frequency.

From RF to Complex Baseband: Spectral Interpretation

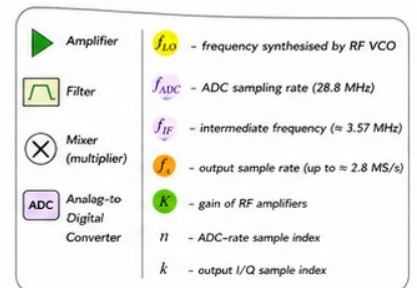


The receiver selects the desired RF channel centered at f_c with bandwidth B_{RF} .

The selected RF channel is translated to the low IF $f_{IF} = |f_c - f_{LO}| \approx 3.57 \text{ MHz}$. The real sampled signal has a conjugate-symmetric spectrum (images at $\pm f_{IF}$) within $[-f_{ADC}/2, f_{ADC}/2]$.

The signal is translated to baseband (centered at 0 Hz), represented as a complex signal. The output bandwidth is B_{BB} and negative frequencies do not contain redundant information.

The SDR does not demodulate the information-bearing signal: it translates the selected RF channel into a sampled complex-baseband representation.



RTL-SDR Signal-Processing Steps

1. The R820T tuner selects the desired RF band and translates it to a low intermediate frequency, approximately $f_{IF} = 3.57$ MHz.
2. The analog IF signal is filtered and sampled by an 8-bit ADC clocked at

$$f_{ADC} = 28.8 \text{ MHz}$$

3. The resulting real digital IF samples $v_{IF}[n]$ are mixed with quadrature NCO signals, low-pass filtered and decimated to the user-selected output sample rate f_s . The sample rate is configurable up to approximately $f_s = 2.8$ MS/s.
4. The down-converted complex samples are then passed to the host computer over a standard USB connection.

From Complex I/Q Samples to an FM Receiver

- The RTL-SDR provides a sequence of complex baseband samples

$$\tilde{v}[k] = v_I[k] + jv_Q[k],$$

which preserves the amplitude and phase evolution of the selected RF channel.

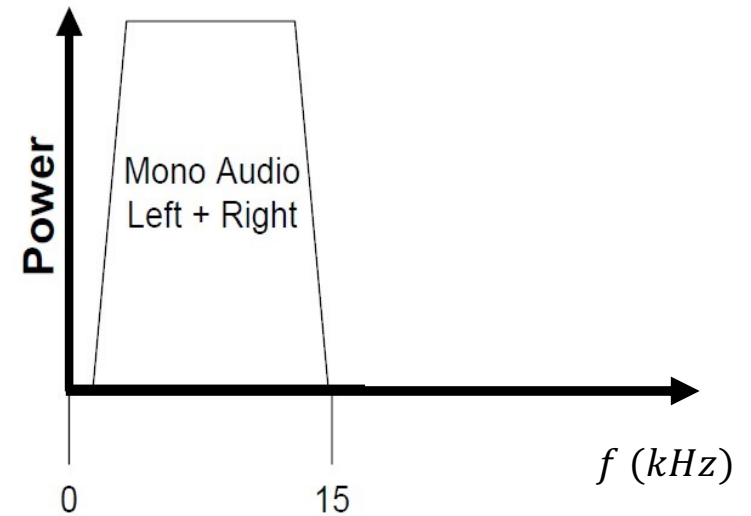
- To recover the transmitted information, we must now apply a modulation-specific baseband receiver.
- The SDR performs RF acquisition and conversion to complex baseband; FM demodulation is implemented afterwards in software in MATLAB.

FM Mono Broadcasting Signal

- Mono FM broadcasting was introduced around 1945 and quickly became a major commercial success.
- The modulating signal $m(t)$ is obtained by combining the left $L(t)$ and right $R(t)$ audio channels

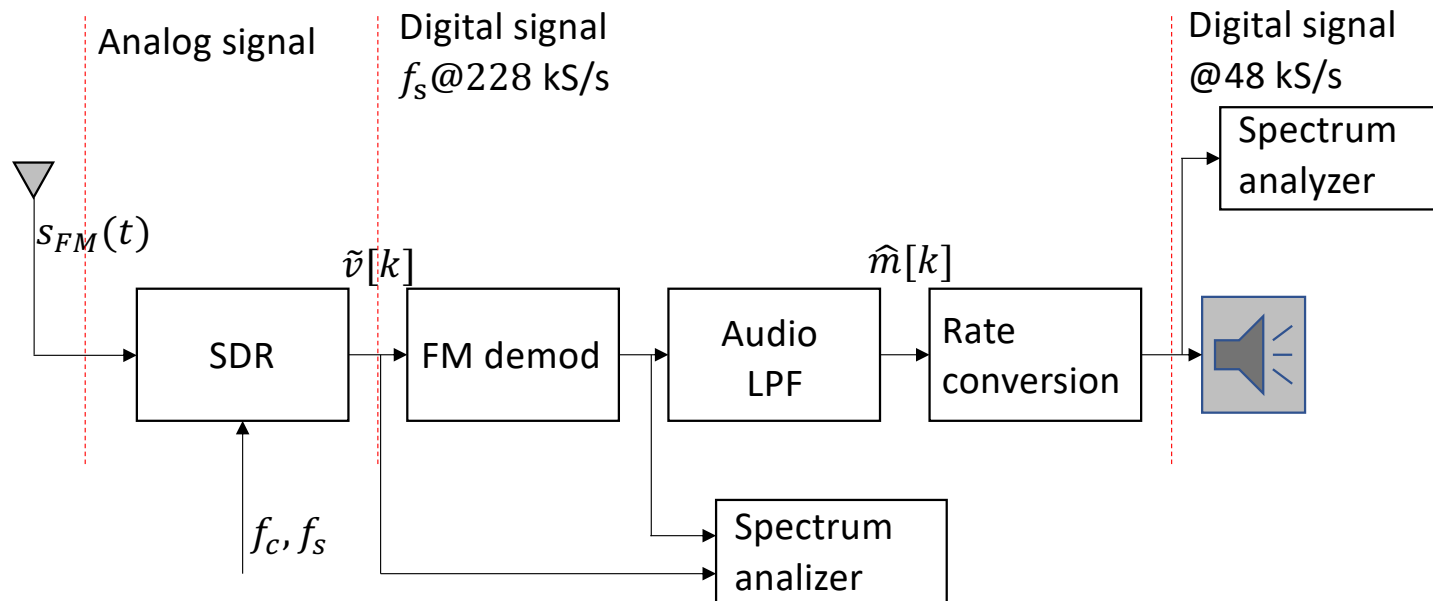
$$m(t) = L(t) + R(t)$$

- The message $m(t)$ is
 - low-pass filtered to a bandwidth $B = 15$ kHz,
 - normalized to 1 and $k_f = 75$ kHz/V so that $\Delta f = k_f \max_t |m(t)| = 75$ kHz



$$m_f = \frac{75}{15} = 5$$

FM Reception with the RTL-SDR



From FM Phase to the Message Signal

- Neglecting the effect of noise and channel, the complex envelope of the received signal is

$$\tilde{v}(t) = A_c e^{j2\pi k_f \int_{-\infty}^t m(\tau) d\tau}$$

- The modulating signal can be recovered by differentiating the phase of $\tilde{v}(t)$

$$\hat{m}(t) = \frac{1}{2\pi k_f} \frac{d}{dt} \angle \tilde{v}(t)$$

- The SDR does not provide $\tilde{v}(t)$, but its samples $\tilde{v}[k]$. We therefore need a discrete-time approximation of the phase derivative.

Discrete-Time FM Discriminator

- At the SDR output there are the samples of the signal complex envelope taken at multiples of $T_s = 1/f_s$. Neglecting the noise we have

$$\tilde{v}[k] = \tilde{v}(kT_s) = A_c e^{j2\pi k_f \int_{-\infty}^{kT_s} m(\tau) d\tau}$$

- The product of two consecutive baseband samples yields

$$\tilde{v}[k] \tilde{v}^*[k-1] = A_c^2 e^{j2\pi k_f \int_{(k-1)T_s}^{kT_s} m(\tau) d\tau} \approx A_c^2 e^{j2\pi k_f m[k] T_s}$$

- An estimate of $m[k]$ is

$$\hat{m}[k] \approx \frac{1}{2\pi k_f T_s} \angle \tilde{v}[k] \tilde{v}^*[k-1] = \frac{1}{2\pi} \frac{f_s}{\Delta f} \angle \tilde{v}[k] \tilde{v}^*[k-1]$$

- Multiplying the current sample by the complex conjugate of the previous sample removes the accumulated phase and retains only the *phase increment* between consecutive samples.